

Imad EL BADISY

Computational Methodologist

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PROFILE

Computational methodologist, PhD, building software for machine learning and computational biostatistics. My methodological research centers on scientific machine learning, survival analysis, missing data imputation, and agentic data science. I hold a PhD in Biomedical Sciences from Aix-Marseille University (SESSTIM laboratory) on missing data and machine learning methods for survival analysis. I turn advanced statistical and machine learning methods into reliable, open-source R packages, including tools for survival machine learning, neural survival models, and imputation, and I contribute as a consultant and trainer in biostatistics and machine learning.

SKILLS

Methods: Scientific machine learning · Survival analysis · Missing data imputation · Agentic data science · Computational biostatistics · Biostatistics and epidemiology · Meta-analysis and evidence synthesis · Interpretable machine learning · Deep learning · Monte Carlo simulation

Tools: R package development · Python · Quarto/LaTeX · Git/GitHub · Slurm/HPC · Large-scale administrative health data pipelines (SNDS)

Languages: Arabic (native) · French (fluent) · English (fluent)

EXPERIENCE

Mohammed VI Center for Research and Innovation (CM6RI)

Rabat, Morocco

Head of AI and Data Science Service

Jan. 2026 – Present

- Leading the AI and Data Science service
- Conducting methodological and applied research in health (oncology, fertility, other biomedical domains)
- Providing methodological support to researchers in computational biostatistics, statistical modeling, and machine learning
- Developing reproducible workflows for health data science and biomedical research projects

Mohammed VI Center for Research and Innovation (CM6RI)

Rabat, Morocco

Research Engineer in Health Data Science

Nov. 2021 – Jan. 2026

- Conducted methodological and applied research in health (oncology, fertility, other biomedical domains)
- Provided methodological support to researchers in computational biostatistics, statistical modeling, and machine learning
- Developed reproducible workflows for health data science and biomedical research projects

The GRAPH Network

Switzerland / Remote

Data Science Education Officer

Sep. 2023 – Dec. 2023

- Designed and delivered training programs in data science for global health
- Developed educational material and course content for data science learners
- Contributed to capacity building in applied data science within a public health context

Institut Mines-Télécom, IMT Atlantique

Brest, France

Research and Development Engineer

Sep. 2019 – Sep. 2021

- Conducted statistical analyses of national health data using SNDS-certified workflows at scale
- Implemented automated, reproducible workflows for complex administrative health data
- Supported applied research projects involving health data, epidemiology, and statistical modeling

INSERM

Clermont-Ferrand, France

Biostatistician

Jan. 2019 – Jun. 2019

- Conducted cost-effectiveness evaluation of a clinical protocol on ketamine for chronic pain
- Contributed to statistical and health-economic analyses in a clinical research setting

Université de Sherbrooke

Sherbrooke, Quebec, Canada

Health Economist Intern

Feb. 2017 – Sep. 2017

- Conducted a systematic review and meta-analysis on health utilities in cancer patients
- Contributed to evidence synthesis and health economics research

TEACHING

Mohammed VI Center for Research and Innovation (CM6RI) <i>Trainer, Machine Learning for Health</i> Applied machine learning for health data analysis · Courses delivered to PhD students	Rabat, Morocco 2026
Aix-Marseille University, SESSTIM <i>Mentor, Artificial Intelligence for Public Health</i> Courses delivered to Master's students · Machine learning methods · Natural language processing	Marseille, France 2022 – Present
University Mohammed VI of Health Sciences (UM6SS) <i>Lecturer, Health Data Science</i> Course delivered to undergraduate health students	Casablanca, Morocco 2021 – Present
Institut Mines-Télécom, IMT Atlantique <i>Lecturer, Health Data Analysis and Decision Support</i> Course delivered to engineering students	Brest, France 2020 – Present

Outside working hours: Organizer of Methods in Health Data Science with R (MHDSR), an independent training pathway (BIOSTATR, ML4HOR) covering applied biostatistics, survival analysis, and machine learning with R

EDUCATION

Aix-Marseille University, SESSTIM <i>PhD in Biomedical Sciences</i> Thesis: Missing Data and Machine Learning Methods for Survival Analysis · Supervisor: Pr Roch Giorgi	Marseille, France 2021 – 2025
Aix-Marseille University, SESSTIM <i>Master 2 in Quantitative Methods for Health Research</i>	Marseille, France 2018 – 2019
Université Clermont Auvergne <i>Master 1 in Applied Mathematics and Statistics</i>	Clermont-Ferrand, France 2017 – 2018
Université Clermont Auvergne, CERDI <i>Master 2 in Health Economics</i>	Clermont-Ferrand, France 2017

SOFTWARE (R PACKAGES)

survalis: Interpretable Survival Machine Learning Framework 19 Survival learners, 7 evaluation metrics, 8 interpretability methods · Maintainer	CRAN · v0.7.1
funcml: Functional Machine Learning Framework Unified framework for functional machine learning in R · Maintainer	CRAN · v0.7.1
survdsn: Deep Neural Networks for Survival Analysis (R torch) Cox, AFT, and discrete-time neural survival models · Maintainer	CRAN · v0.7.6
tvrnst: Time-Varying RMST from Survival Matrices Dynamic RMST summaries for individualized survival curves · Maintainer	CRAN · v0.0.6
unsurv: Unsupervised Clustering of Individualized Survival Curves Phenotyping patients from predicted survival curves · Maintainer	CRAN · v0.5.0
mcstatsim: Monte Carlo Statistical Simulation Tools Functional approach to monte carlo simulation studies in R · Maintainer	CRAN · v0.5.0
missCforest: Ensemble Conditional Trees for Missing Data Imputation Missing data imputation using conditional forest ensembles · Maintainer	CRAN · v0.0.8
functionals: Functional Programming with Parallelism and Progress Tracking Functional programming utilities with parallelism and progress tracking · Maintainer	CRAN · v0.5.0

mimar : Compact Multiple Imputation, Assessment, and Reporting	CRAN · v0.8.0
• Streamlined multiple imputation workflow with assessment and reporting · Maintainer	
testflow : Statistical Testing, Interpretation, and 'ggplot2'-Based Visualization	CRAN · v0.9.0
• Workflow for statistical testing, interpretation, and visualization · Maintainer	
densempl : Dense Neural Networks for Tabular Classification and Regression	CRAN · v0.5.0
• Dense neural network models for tabular data · Maintainer	
CEACT : Cost-Effectiveness Analysis Toolkit for Clinical Trials	CRAN · v0.5.0
• Cost-effectiveness analysis toolkit for clinical trial data · Maintainer	
missknn : Fast Masked K-Nearest Neighbor Imputation	CRAN · v1.0.0
• Fast masked k-nearest neighbor imputation · Maintainer	

SELECTED TALKS

Comparison of missing data imputation methods in Cox regression with time-varying effects <i>ISCB 2023, Rome</i> · El Badisy I. et al.	2023
Missing data imputation using a meta-algorithm metaCART: simulation under MCAR mechanism <i>EpiClin 2023, Nancy</i> · El Badisy I. et al.	2023

PUBLICATIONS

1. **El Badisy I.** (2026). unsurv: Clustering Individualized Survival Curves. *Bioinformatics Advances*, vbag218.
2. **El Badisy I.** (2026). The R Journal: SurvDNN: Survival Deep Learning Models for Tabular Data. *The R Journal*, 18(1):384-399. <https://doi.org/10.32614/RJ-2026-008>.
3. **El Badisy I.** (2026). funcml: Functional Machine Learning Software for R. *Zenodo*. <https://doi.org/10.5281/zenodo.20707605>.
4. **El Badisy I.**, Assarag B., Belrhiti Z. (2026). Interpretable clinical decision support systems in high-risk pregnancy: a scoping review of models, methods, and implementation. *BMC Pregnancy and Childbirth*, 26(1):125.
5. Bnimoussa J., Bouaddi O., **El Badisy I.**, Khalis M. (2026). Drivers of youth mental health and wellbeing: a large-scale cross-sectional study in Morocco. *BMJ Open*, 16(6):e110683.
6. Lamchabbek N., Huybrechts I., **El Badisy I.**, et al. (2026). Global assessment of the relationship between breast cancer risk and dietary intake: an ecological study. *American Journal of Health Promotion*, 40(5):591-600.
7. Bouaddi O., **El Badisy I.**, Charaka H., et al. (2026). Leveraging the Positive Deviance Approach to Drive Behavior Change in Noncommunicable Diseases: A Scoping Review. *Public Health Challenges*, 5(1):e70074.
8. Lamchabbek N., Mrah S., Mane N., **El Badisy I.**, et al. (2026). Adherence to WCRF/AICR cancer prevention recommendations and reduction in breast cancer risk: results of a large-scale case-control study in Morocco. *The British Journal of Nutrition*, 1-29.
9. Houdou A., Khomsi K., Delle Monache L., **El Badisy I.**, et al. (2025). Enhancing 5-day particulate matter (PM10) forecasts in Morocco using U-net: A deep learning approach. *Atmospheric Research*, 108439.
10. Kadi C., Ahmadi N., Houdou A., **El Badisy I.**, et al. (2025). Differentiating Latent Tuberculosis from Active Tuberculosis Through Activation Phenotypes and Chemokine Markers HLA-DR, CD38, MCP-1, and RANTES: A Systematic Review and Meta-Analysis. *Biomarker Insights*, 20:11772719241312776.
11. Lyubareva I., Rochelandet F., **El Badisy I.** (2025). Imitation of Business Models: The Case of Online News Industry in France. *International Journal of Arts Management*, 27(2):67-85.
12. Houdou A., **El Badisy I.**, Khomsi K., et al. (2024). Interpretable machine learning approaches for forecasting and predicting air pollution: A systematic review. *Aerosol and Air Quality Research*, 24(1):230151.
13. **El Badisy I.**, BenBrahim Z., Khalis M., et al. (2024). Risk factors affecting patient survival with colorectal cancer in Morocco: survival analysis using an interpretable machine learning approach. *Scientific Reports*, 14(1):3556.
14. **El Badisy I.**, BenBrahim Z., Khalis M., et al. (2024). Author Correction: Risk factors affecting patient survival with colorectal cancer in Morocco: survival analysis using an interpretable machine learning approach. *Scientific Reports*, 14:9985.

15. **El Badisy I.**, Graffeo N., Khalis M., Giorgi R. (2024). Multi-metric comparison of machine learning imputation methods with application to breast cancer survival. *BMC Medical Research Methodology*, 24(1):191.
16. Zbiri S., Belghiti Alaoui A., **El Badisy I.**, et al. (2024). Private hospitals in low-and middle-income countries: a typology using the cluster method, the case of Morocco. *BMC Health Services Research*, 24(1):1231.
17. Abdala S.A., Khomsi K., Houdou A., **El Badisy I.**, et al. (2024). Emission reduction strategies and health: a systematic review on the tools and methods to assess co-benefits. *BMJ Open*, 14(12):e083214.
18. **El Badisy I.**, Nejari C., Naim A., et al. (2023). CO10. 6-Imputation des données manquantes par un méta-algorithme (metaCART): étude de simulation. *Epidemiology and Public Health = Revue d'Epidémiologie et de Santé Publique*, 71:101632.
19. Jallal M., Berrada K., Bouaddi O., **El Badisy I.**, et al. (2023). How physicians in a Moroccan tertiary care center perceive teleconsultation during COVID-19 pandemic? *Telemedicine and e-Health*, 29(2):284-292.
20. Khalis M., Tembo J.M.W., Elmouden L., **El Badisy I.**, et al. (2023). Tobacco Use Among Dental Students in Morocco: Opportunities for Professional Cancer Education. *Journal of Cancer Education*, 38(3):821-828.
21. Houdou A., Khomsi K., Leghrib R., **El Badisy I.**, Abdala S.A., Ibrahimi A., et al. (2023). Forecasting Particulate Matter (PM10) Using Machine Learning Algorithms in Agadir. *ISEE Conference Abstracts*, 2023(1).
22. Houdou A., **El Badisy I.**, Abdala S.A., Khomsi K., Khalis M. (2023). A Machine Learning Approach for Particulate Dust Forecasting in Morocco. *ISEE Conference Abstracts*, 2023(1).
23. Khalis M., Toure A.B., **El Badisy I.**, et al. (2022). Relationship between meteorological and air quality parameters and COVID-19 in Casablanca region, Morocco. *International Journal of Environmental Research and Public Health*, 19(9):4989.
24. Lyubareva I., Rochelandet F., **El Badisy I.** (2022). Evolution of business models and industrial organization. Uncertainty, competition and imitation behavior in the French Press Industry. *16th International Conference on Arts and Cultural Management*.
25. Le Goff-Pronost M., **El Badisy I.**, Dejonghe Q., et al. (2022). 5 years of early management of skin cancer by tele-dermatology: what have we learned from real-world data? *International Conferences on E-Society 2022 and Mobile Learning 2022*, 85.
26. Lyubareva I., Mesangeau J., Boudjani N., **El Badisy I.**, Brisson L. (2021). La plateforme des médias français et le ton du débat public. Exemple de YouTube. *Communication. Information médias théories pratiques*, 38(2).
27. Le Stum M., Dardenne G., **El Badisy I.**, et al. (2021). Dynamiques des poses de Prothèse de Genou en France. Projection à 2030. *Colloque annuel de l'Institut Brestois en biologie Santé Matière (IBSAM)*.